

run_q3

- **reduction**

- **lower**

- Matrix.hstack

- sp.simplify

- exact_integer_quotient

- **recurrence normalization**

- **cancel3 → normalize3**

- lcm_expr / sp.lcm

- fraction denominator

- sp.together

- sp.expand

- sp.gcd

- math.gcd

- Poly.LC

- sp.factor

- **coefficient checks**

- Python sum

- Poly.nth

- **differential equation**

- **make_ode → apply2**

- coefficients of A'' , A' , A

- ODE residual at $A=S$